Chris Toomey, Jenice An, and Tongyu Guo
ADAN 8888 - Applied Analytics Project - Spring 2026
Module 6 Assignment - Develop First Modeling Approach

HelpHerInvest: A machine learning-based investment recommendation system that aligns user interests and risk preferences with benchmark stock evaluation.

Modeling Approach

NLP BERT Model

Our group applied Term Frequency-Inverse Document Frequency (TF-IDF) to convert the textual data into an embedded matrix. Essentially this text preprocessing aims to tokenize text, remove stop words, and compute weights for machine learning models. This allows the model to understand patterns within the text and ultimately make predictions based on textual data. Next steps were to utilize a Bidirectional Encoder Representations from Transformers (BERT) model, which was pre-trained and can be easily implemented on embedded data. This allows our group to perform semantic analysis on the stock tickers based on the company's description, section, or industry. The end goal is to have this model generate a subset of tickers based on a user's input and how similar the description is. This will allow us to then apply the classification model on the historical financial data on these tickers to determine which may have the best probability of outperforming the S&P 500.

Logistic regression model

For our first modeling approach, we've selected logistic regression since our objective is to classify stocks based on their forward excess return relative to the S&P 500, which makes this a supervised classification problem rather than a continuous forecasting task. Logistic regression is well suited because it provides us a structured way to examine how financial indicators relate to future performance outcomes. As a first step, it allows us to test whether our features have predictive value. By starting with a simpler model, we are able to establish a baseline level of performance before exploring more complex techniques.

Our group implemented the following three variations of a logistic regression model:

1. L1 (Lasso) Logistic Regression

L1 Lasso logistic regression adds an L1 penalty, which provides sparsity among the features. This often drives coefficients of less meaningful features to zero, essentially performing feature

selection within the model. The goal is to have this improve model interpretability. In our case, it did not appear to eliminate any features.

2. Elastic Net Logistic Regression

Elastic net logistic regression utilizes L1 (Lasso) and L2 (Ridge) regularization penalties to assist in dealing with multicollinearity among the features. It applies sparsity among the features, but also utilizes stability of correlated features. We received similar results when it came to the coefficients and classification report metric (accuracy, recall, F1-score, etc.) to the L1 Lasso model.

3. Weighted Logistic Regression

The weighted logistic regression does not change the regularization, compared to the other two models. The weighted logistic regression model assigns specific weights to observations, which can allow certain features to influence the model more than others. Overall, this performed the best when it came to the classification report metrics.

Complexity of the Modeling Approach

Logistic regression represents a relatively simple and controlled modeling approach for our first attempt at making predictions on this dataset. We went with three variations using different parameter settings to better understand how the model behaves under different regularization strategies. Because this is our initial run, we intentionally chose a method that is straightforward and will provide interpretable insight into how our features relate to our target variable (forward excess return).

Hyperparameters

1. L1 (Lasso) Logistic Regression

We chose to use the L1 (Lasso) penalty to add some form of regularization on the features as well as the saga solver because it tends to do well with larger datasets. This is a relatively simple hyperparameter selection and we viewed it as a potential baseline to see how the model fair with minimal influence.

2. Elastic Net Logistic Regression

We utilized elastic net penalty for the second model, which utilizes both the L1 (Lasso) and L2 (Ridge) regularization. Similar to before, this is a simple approach to handle feature sparsity but can also apply some stability to the coefficients.

3. Weighted Logistic Regression

Lastly, we chose a class weight balanced approach for the final model. THis is typically useful when it comes to dealing with an imbalanced dataset. Though our dataset was not overly imbalanced when it came to the forward excess return being positive (143,948 observations) or negative (151,730 observations), the predictions of the first two models were. Ultimately, the weighted logistic regression hyperparameter produced the best results.

Model Performance Metrics

Reviewing the classification report in sklearn, it provides you with useful metrics (such as accuracy, precision, recall, and f1-score) to help quantify how your model is performing. Accuracy measures the overall correctness of the model, whereas precision represents the ratio of correctly predicted positive observations to the total predicted positives. Recall is similar, but the ratio of correctly predicted positive observations to all actual positives - telling you how many of the actual positives did the model identify. Lastly, F1 score is a blend between the precision and recall that captures both under one metric. We decided to focus on the F1 score to provide that blended metric.

Validation Metric Values for all 3 Variations

L1 Logistic Regression			
	precision	**recall**	**f1-score**
0	0.5316	0.8549	0.6556
1	0.5004	0.1618	0.2445
accuracy	0.5269	0.5269	0.5269
macro avg	0.5160	0.5083	**0.4500**

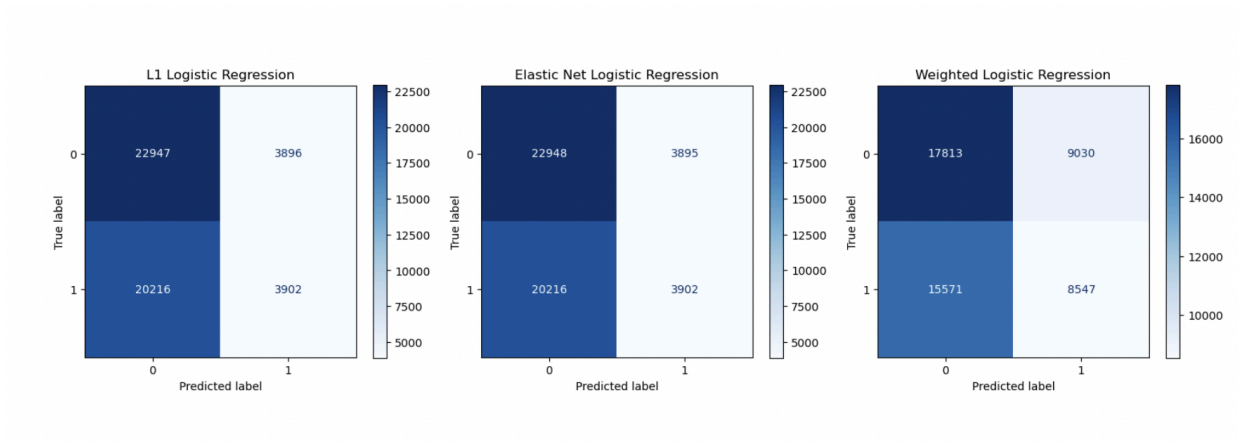
weighted avg	0.5168	0.5269	0.4610

Elastic Net Logistic Regression			
	precision	**recall**	**f1-score**
0	0.5316	0.8549	0.6556
1	0.5004	0.1618	0.2445
accuracy	0.5269	0.5269	0.5269
macro avg	0.5160	0.5083	**0.4501**
weighted avg	0.5169	0.5269	0.4610

Weighted Logistic Regression			
	precision	**recall**	**f1-score**
0	0.5336	0.6636	0.5915
1	0.4863	0.3544	0.4100
accuracy	0.5173	0.5173	0.5173
macro avg	0.5099	0.5090	**0.5008**
weighted avg	0.5112	0.5173	0.5056

The L1 Lasso logistic regression and the elastic net logistic regression had very similar metric results. The precision, recall, and F1-scores for both of these models were nearly identical, aside from a slight variation in the macro average F1-score. The weighted logistic regression had a better overall performance and showed a more diverse matrix of predictions. Overall, both models heavily favor the 0 class (forward excess return underperforming the benchmark). This

is certainly something we will need to address in the coming models and we will likely have to perform feature engineering / hyperparameter tuning to overcome.



Winning Model for Week 6

Our strongest performing model was the weighted logistic regression model. The difference between the validation and testing sets was relatively similar, which is a positive. This indicates that the model is not overfitting and can generalize the data fairly well, but we would still prefer the performance of the model to improve.

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Weighted Logistic Regression Performance - Test Set:
Classification Report:

              precision    recall  f1-score   support

           0       0.55      0.71      0.62     25892
           1       0.50      0.34      0.40     22601

    accuracy                           0.54     48493
   macro avg       0.53      0.52      0.51     48493
weighted avg       0.53      0.54      0.52     48493

F1: 0.512
ROC-AUC: 0.5351
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